Name: ____________

Date: ____________

Function Combinations & Polynomials

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce | D — Deepen | A — Apply | R — Reinforce

☐ **1. I can add, subtract, multiply, and divide functions represented by tables, graphs, or formulas.**

RELATED SKILLS

- I — Add, subtract, multiply, and divide function outputs.
- A — Identify input and output values in a formula, graph, table, or context.

☐ **2. I can determine where a combination of functions is positive, zero, or undefined.**

RELATED SKILLS

- I — Determine the domain of a sum, difference, product, quotient, or composition.
- D — Add, subtract, multiply, and divide function outputs.

☐ **3. I can determine polynomial end behavior from degree and leading coefficient.**

RELATED SKILLS

- I — Infer polynomial end behavior from degree and leading coefficient.
- A — Identify terms, factors, coefficients, variables, and degree in a polynomial expression.

☐ **4. I can identify zeros and multiplicities and relate them to polynomial graph behavior.**

RELATED SKILLS

- I — Relate zero multiplicity to local graph behavior.
- D — Determine real and complex zeros using an appropriate method.

☐ **5. I can estimate a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.**

RELATED SKILLS

- I — Use inflection points to establish a minimum possible polynomial degree.
- I — Use turning points to establish a minimum possible polynomial degree.
- I — Use zeros and their multiplicities to establish a minimum possible polynomial degree.
- D — Infer polynomial end behavior from degree and leading coefficient.
- D — Relate zero multiplicity to local graph behavior.

☐ **6. I can construct a polynomial function from specified zeros, multiplicities, and a point.**

RELATED SKILLS

- D — Build a polynomial from zeros, multiplicities, and a point.
- A — Determine real and complex zeros using an appropriate method.
- A — Relate zero multiplicity to local graph behavior.

7. I can match polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.

RELATED SKILLS

- D Infer polynomial end behavior from degree and leading coefficient.
- A Determine real and complex zeros using an appropriate method.
- A Relate zero multiplicity to local graph behavior.

8. I can build and optimize a polynomial model in a geometric or economic context.

RELATED SKILLS

- D Interpret vertex and zeros in an application.
- A Explain a model parameter in the language of its context.
- A Restrict a model to meaningful contextual inputs.